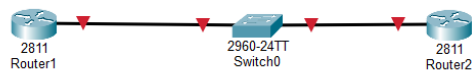


WEEK5 PRAC, NET512

Refiloe Seselinyane

30 August 2026

20260711



28) R1:

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

29) R2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface f0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

30) SW1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
```

31)

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

32)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface f0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

33)

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#
% Invalid input detected at '^' marker.

Switch(config)#interface vlan 1
Switch(config-if)#ip address 10.10.10.10 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#ip default-gateway 10.10.10.2
Switch(config)#interface f0/1
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface f0/2
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

34) Ping R2 on SW1

```
Switch#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/2 ms
```

IP INTERFACE BRIEF:

```
Switch#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/2 unassigned      YES manual up          up
FastEthernet0/3 unassigned      YES manual down        down
FastEthernet0/4 unassigned      YES manual down        down
FastEthernet0/5 unassigned      YES manual down        down
FastEthernet0/6 unassigned      YES manual down        down
FastEthernet0/7 unassigned      YES manual down        down
FastEthernet0/8 unassigned      YES manual down        down
FastEthernet0/9 unassigned      YES manual down        down
FastEthernet0/10 unassigned      YES manual down        down
FastEthernet0/11 unassigned      YES manual down        down
FastEthernet0/12 unassigned      YES manual down        down
FastEthernet0/13 unassigned      YES manual down        down
FastEthernet0/14 unassigned      YES manual down        down
FastEthernet0/15 unassigned      YES manual down        down
FastEthernet0/16 unassigned      YES manual down        down
FastEthernet0/17 unassigned      YES manual down        down
FastEthernet0/18 unassigned      YES manual down        down
FastEthernet0/19 unassigned      YES manual down        down
FastEthernet0/20 unassigned      YES manual down        down
FastEthernet0/21 unassigned      YES manual down        down
--More--
```

35)

36)

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#
```

37)

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme    Capability  Platform  Port ID
R1#
```

38)

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
|
```

39)

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R1#
```

40)

```
R2>ena
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.10.10.2      YES manual up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#
```

41)

```
R2(config)#int F0/0
R2(config-if)#shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

R2(config-if)#exit
R2(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.10.10.2      YES manual administratively down down
FastEthernet0/1 unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#
```

42)

```
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.10.10.2      YES manual up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#
```

43)

```
Switch(config-if)#duplex half
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface f0/1
FastEthernet0/1 is up, line protocol is up (connected)
  Hardware is Lance, address is 0090.2b72.9201 (bia 0090.2b72.9201)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
--More--
```

44)

```
Switch#show interface f0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0090.2b72.9202 (bia 0090.2b72.9202)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
--More--
```

45)

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/2
Switch(config-if)#speed 10
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces f0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0090.2b72.9202 (bia 0090.2b72.9202)
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue : 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
  2357 packets output, 263570 bytes, 0 underruns
--More--
```

46)

```
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    10.10.10.2      YES manual up              up
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2#
```

47) Interface stays up/up but with high collisions, late collisions, CRC/FCS errors, very slow throughput, ping loss or high latency, and show interfaces shows input errors increasing.